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Earth and Planetary Science Letters 194 (2001) 135–149

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Paleomagnetism of Carboniferous sediments in the Hexi corridor: its origin and tectonic implications

Baochun Huang [a,*], Yo-ichiro Otofuji [b], Rixiang Zhu [a], Ruiping Shi [a], Yongcheng Wang [a]

[a] *Institute of Geology and Geophysics, Chinese Academy of Sciences, P.O. Box 9701, Beijing 100101, PR China*
[b] *Department of Earth and Planetary Sciences, Faculty of Science, Kobe University, Kobe 657-8501, Japan*

Received 27 May 2001; received in revised form 20 September 2001; accepted 13 October 2001

Abstract

We report paleomagnetic results for Carboniferous rocks from the eastern part of the Hexi Corridor, which we have demonstrated to be the western extension of the North China block (NCB). The characteristic high-temperature remanence component (B-component) exhibits predominantly reverse polarity, and is generally separable by thermal demagnetization temperature steps between 350 and 580°C. This B-component can also be isolated by alternating fields (AFs) cleaning at a peak value of 20–30 mT using a hybrid thermal–AF demagnetization method. The B-component shows a positive fold test at the 95% confidence level, indicating that it was acquired before Late Triassic to Early Jurassic folding. We interpret the B-component to be of primary origin acquired during the Carboniferous. The corresponding pole positions are at 14.0°E, 10.5°N, $\alpha_{95} = 6.2°$ for the Early-Middle Carboniferous (late Visean to Namurian) and at 10.2°E, 33.3°N, $\alpha_{95} = 16.7°$ for the Late Carboniferous (Stephanian). The preliminary Late Carboniferous pole is in agreement with that for the NCB, supporting earlier claims that the Hexi Corridor has been the western extension of the NCB since the Ordovician. On the other hand, these two Carboniferous paleopoles are distinct from the coeval ones for Inner Mongolia, Outer Mongolia, South China, Tarim and stable Europe at greater than 95% confidence, suggesting that these continental blocks could not have been connected in their present configuration during Carboniferous time. The significant difference in paleolatitude between the NCB and Australian part of Gondwanaland according to the two new Hexi Corridor poles indicates that they had drifted apart by the late Visean. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Ningxia China; North China Platform; Carboniferous; paleomagnetism

1. Introduction

The Phanerozoic apparent polar wander (APW) paths for three major continental blocks of China (North China, South China and Tarim) have been recently revised based upon a significant amount of new paleomagnetic results [1–4]. Earlier published APW paths for the three major continental blocks of China [5–9] were revised on the bases of newly available paleomagnetic poles for the Early Paleozoic resulting in better under-

* Corresponding author. Tel.: +86-10-6488-9024; Fax: +86-10-6487-1995.
E-mail address: bchuang@mail.c-geos.ac.cn (B. Huang).

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PII: S0012-821X(01)00557-X

standing of the paleogeography and kinematic history during this time [1,2,10–13]. Paleogeography and tectonics during the mid-Paleozoic, however, have remained unresolved problems for the three major continental blocks of China owing to the serious lack of paleomagnetic data for the Middle and Late Paleozoic. For the North China block (NCB) this lack is due in part to the weak magnetic remanence of Upper Carboniferous to Middle Permian sedimentary rocks, which are generally made up of light colored coarse-grained sandstones, and also to a major stratigraphic hiatus from the Upper Ordovician to the Lower Carboniferous [14].

The relationship between the Hexi Corridor terrane and stable NCB, defined as the eastern part of the NCB beyond the Helanshan fold belt (HFB; Fig. 1a), has been constrained recently by a paleomagnetic study in the eastern part of the Hexi Corridor, Ningxia of North China [15]. This study demonstrated that a paleomagnetic pole for Early-Middle Ordovician limestone rocks from the eastern part of the Hexi Corridor is in good agreement with NCB poles, implying that the Hexi Corridor had been part of the NCB since the Ordovician in its present configuration. This finding paleomagnetically favors that some post-Ordovician Paleozoic lithologies missing on the stable NCB are present in the Hexi Corridor and forms the basis for consideration of post-Ordovician Paleozoic problems of the NCB using Silurian, Devonian and Carboniferous results from the eastern part of the Hexi Corridor terrane, extensively we sampled recently in several sections.

The Miboshan area southwest of Ningxia, where an Early-Middle Ordovician paleomagnetic pole had been obtained [15], is just about 10 km from our Silurian and Devonian sampling sections, the Xiaoyuchuan section, and our Silurian and Devonian results from three separated sampling sections (Zhaohuajing, Shixiagou and Xiaoyuchuan) show good consistency, indicating that no relative rotation took place between them [13]. Thus, our results from the eastern part of the Hexi Corridor for the Ordovician, Silurian and Devonian, may be considered free of local rotation effects and to be representative for the region

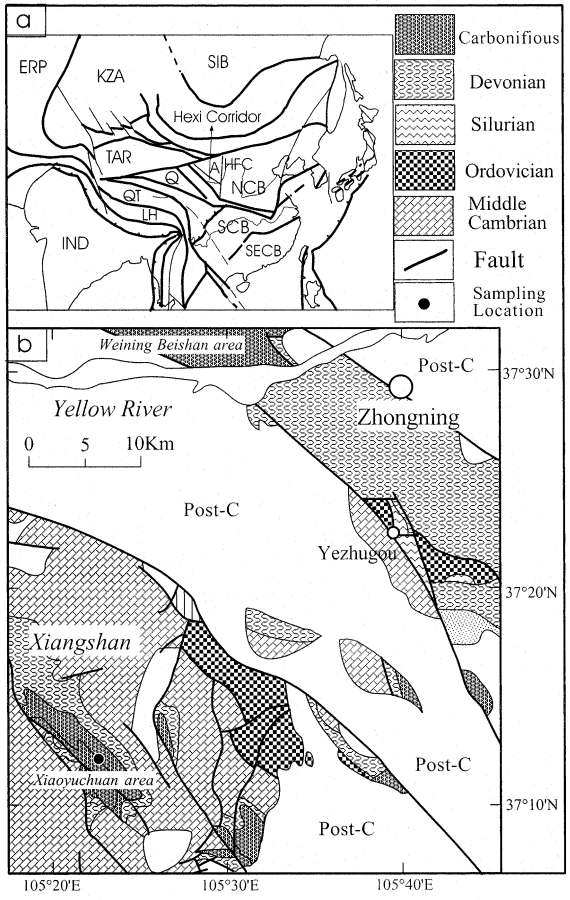


Fig. 1. Sketch maps showing (a) the tectonic framework of southern and eastern Asia (modified after [9]), and (b) the Zhongning area of southwest Ningxia with an overview of Paleozoic geology (modified after [17]). Abbreviations are: A: Alashan; C: Carboniferous; ERP: Europe; HFB: Helanshan fold belt; IND: India; KZA: Kazakhstan; LH: Lhasa block; NCB: North China block; Q: Qaidam block; QT: Qiangtang block; SECB: Southeast China block; SCB: South China block; SIB: Siberia; TAM: Tarim.

and for the NCB. With the aim of better understanding the paleogeography and tectonics of the NCB during the Carboniferous we present paleomagnetic results from the Xiaoyuchuan section of the eastern part of the Hexi Corridor, southwest Ningxia, North China.

We compare the results with the two major continental blocks of China and adjacent areas during the Carboniferous, and present a new paleogeographic reconstruction.

2. Geological setting and paleomagnetic sampling

The Hexi Corridor area is located in the southern margin of the Alashan/Hexi Corridor terrane, to the west of the Ordos Basin across the HFB, and at the juncture of two major Chinese blocks, North China and Tarim (Fig. 1a). Geological evidence and paleomagnetic results strongly suggest that the Alashan/Hexi Corridor terrane has formed the western extension of the NCB since Ordovician time [14–16].

In the eastern part of the Hexi Corridor, the slightly metamorphosed pre-Cambrian basement and Lower Cambrian deposits were mostly eroded by uplift and denudation [13,16]. The Early Paleozoic is composed of the Middle Cambrian Xiangshan Group and the Lower-Middle Ordovician series. They are usually in disconformable or unconformable contact with both overlying and underlying strata. Silurian and Devonian depositions are also well developed in the eastern part of the Hexi Corridor, usually paraconform or unconform contacts [17]. Two major folding events occurred, presumably during the Late Devonian to Early Carboniferous and during the Early Jurassic to Middle Jurassic, corresponding to the Variscan and Indosinian orogenies, respectively. Indeed, two unconformities can be distinguished in the Alashan/Hexi Corridor terrane, the earlier one between Late Devonian and Early Carboniferous sediments, and the later one between Early Jurassic and Middle Jurassic sediments [17].

The Carboniferous system is made up in ascending order of the Lower Carboniferous Chouniugou Formation, Middle Carboniferous Jingyuan and Yanghugou Formations, and the Upper Carboniferous Taiyuan Formation (Fig. 2). The Lower Carboniferous series unconformably overlies the Upper Devonian Zhongning Formation. It is mainly composed of sandy sandstone and mudstone interbedded with limestone or light-gray and gray limestone. The upper part of this series yields corals (e.g. *Yuanophyllum kansuense*, *Dibunophyllum bristolens*), brachiopods (*Gigantoproductus maxima*, *Kansuella kansuensis*) and flora (e.g. *Sublepidodendron mirabile*, *Rhodea hsianghsiangensis*), indicating a late Datang age (i.e. late Visean stage, [17]). The Middle Carboniferous Jingyuan Formation is well exposed in the Xiaoyuchuan section of Ningxia, succeeding without apparent break the underlying Chouniugou Formation in the eastern part of the Hexi Corridor area. It consists largely of grayish black/deep-gray limestone with intercalations of quartz sandstone, yielding ammonites (e.g. *Dombarites paratectus*, *Eumorphoceras chunweiense*) and flora (e.g. *Linopteris* sp., *Paripteris gigantea*), indicating a Namurian age [17]. The Middle Carboniferous Yanghugou Formation overlies the Jingyuan Formation without apparent break, and also consists of sandstone and shale interbedded with limestone. It yields abundant flora (e.g. *Neuropteris gigantea*, *Linopteris neuropteroides*), brachiopods (e.g. *Rhipidomella michelini*, *Chonetes carbonifera*), lamellibranchs (*Posidoniella cf. lavis*, *Dunbarella subpapyracea*) and corals (*Bradyphyllum bellicostatum*), suggesting a Westphalian age [17]. The Upper Carboniferous Taiyuan Formation overlies the Yanghugou Formation without apparent break. The rocks are mainly quartz sandstone, sandy shale and limestone. This series is richly fossiliferous, yielding flora (e.g. *Neuropteris ovata*, *Leppidodendron posthumii*) and brachiopods (e.g. *Chonetes cf. uralicus*, *Dictyoclostus taiyuanfuensis*), indicating a Stephanian age [17].

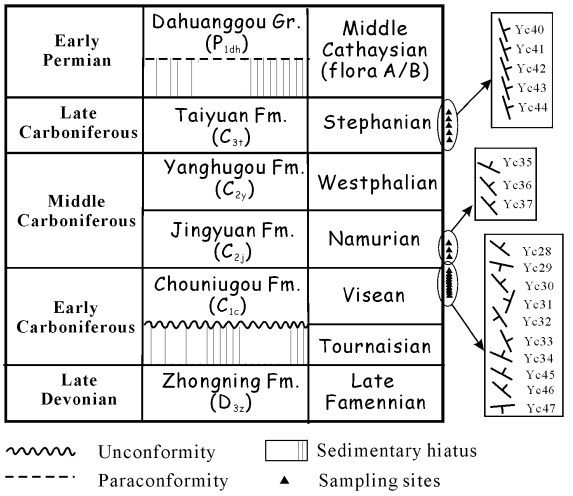


Fig. 2. Sketch plot of a stratigraphic column showing the Carboniferous system in the eastern part of the Hexi Corridor, with sampling sites indicated as black triangles. Strike direction for each sampling site is shown in the insets.

A total of 108 oriented drill-core samples from 18 sites were collected from the Xiaoyuchuan area of Zhongwei County (105.4°E, 37.2°N) in southwest Ningxia of North China where a well-developed Carboniferous sedimentary succession unconformably overlies Upper Devonian or Middle Cambrian series (Fig. 1b). Samples were collected from 10 sites consisting of limestone rocks from the upper part of Early Carboniferous Chouniugou Formation. For the Late Carboniferous, samples were collected from five sites consisting of limestone rocks from the Taiyuan Formation. In addition, samples from two sites consisting of limestone rocks and one site consisting of sandstone rocks were collected from the lower Middle Carboniferous Jingyuan Formation (Fig. 2). Some seven oriented drill-cores distributed stratigraphically over several meters were sampled at each site. All samples were collected with a portable drill and oriented using a magnetic compass. The Late Carboniferous samples were collected from a monoclinal section, whereas the Early-Middle Carboniferous samples were collected from a folded section (Fig. 2 and Table 1).

3. Paleomagnetic results

3.1. Method

All specimens were subjected to either progressive thermal or hybrid thermal–alternating field

Table 1
Summary of sampling data and results for the low-temperature A-component from the Hexi Corridor

Site ID	Strike	Dip	n/n_0	R/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95}
Early–Middle Carboniferous, limestone and one site of sandstone (YC35):										
YC28 (283–291)	128	74	5/5	0/5	14.7	56.9	236.0	45.5	24.3	15.8
YC29 (292–301)	282	83	3/3	0/3	2.0	42.9	2.5	−39.3	11.7	37.8
YC30 (302–311)	139	66	5/5	1/4	353.7	51.8	270.5	39.9	31.6	13.8
YC31 (312–318)	21	20	2/2	0/2	11.3	41.2	29.1	41.5	–	–
YC32 (319–328)	326	20	2/8	0/2	5.7	49.9	18.6	35.3	–	–
YC33 (329–339)	154	37	7/9	0/7	1.5	46.1	317.8	50.2	41.1	9.5
YC34 (340–347)	103.4[a]	45.9[a]	6/7	1/5	12.9	52.2			25.2	13.6
							192.0	81.5	18.7	15.9
YC35 (348–356)	115	26	3/8	3/0	189.6	−18.9	184.6	−43.7	113.5	11.6
YC36 (357–365)	139	28	7/8	0/7	74.7	24.0	85.5	48.2	37.4	10.4
YC37 (366–374)	139	28	5/6	3/2	317.0	−27.4	310.4	−54.4	10.7	24.5
YC45 (445–454)	119	59	7/8	0/7	23.9	37.8	239.5	82.0	21.9	13.2
YC46 (455–464)	131	31	0/5	No low-temperature component isolated						
YC47 (465–473)	265	27	7/7	0/7	11.2	48.7	6.5	22.5	28.1	11.6
Late Carboniferous, limestone:										
YC40 (395–404)	161	22	5/5	0/5	18.4	19.9	9.3	32.0	95.8	7.8
YC41 (405–414)	161	22	0/7	No low-temperature component isolated						
YC42 (415–424)	161	23	5/6	0/5	41.8	38.9	26.1	57.5	29.3	14.4
YC43 (425–433)	163	21	2/2	0/2	69.9	39.3	68.3	60.2	–	–
YC44 (434–444)	163	21	5/7	0/5	359.7	40.9	340.4	43.6	14.1	21.1
Averaging by sites			16/18		21.3	42.3			14.1	10.2
							3.9	57.8	3.9	21.6
Averaging by specimens			76/96	8/68	22.1	42.4			10.0	5.4
							358.2	62.3	3.8	9.7

Abbreviations are: site ID, site identification with sample number range showing between parentheses; strike and dip, strike azimuth and dip of bed; n/n_0, number of specimens used in the calculation/number of specimens demagnetized; R/N, specimens with reverse/normal polarity; D_g, I_g (D_s, I_s), declination and inclination of direction in-situ (after tilt correction).

[a] Mean value (YC34a (340–341), strike/dip = 120/48; YC34b (342–347), strike/dip = 90/47).

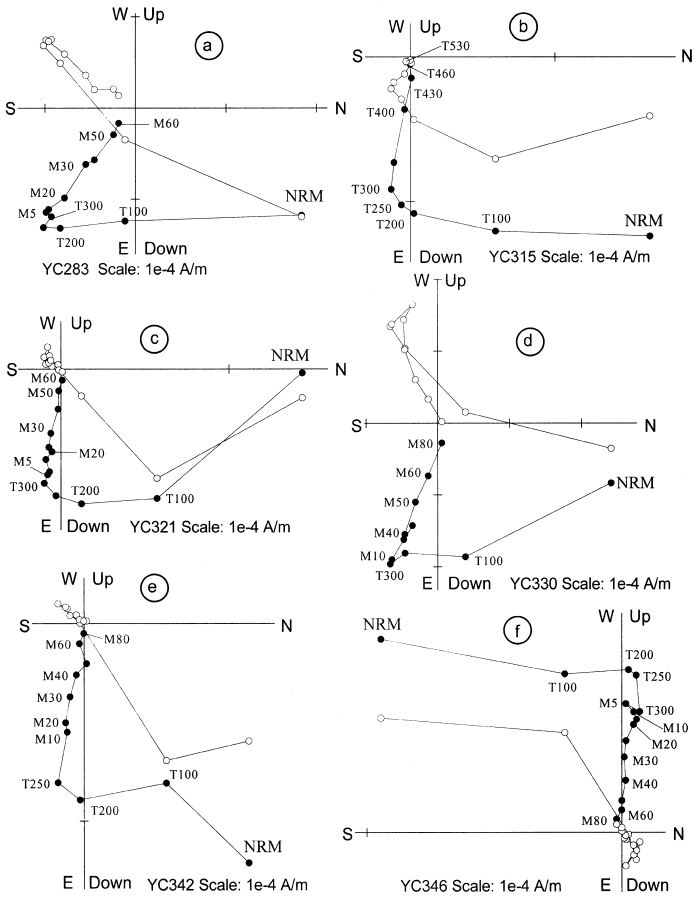


Fig. 3. Typical orthogonal vector plots for the Early-Middle (a–g) and Late (h–j) Carboniferous specimens. Directions are plotted in-situ, open and solid circles represent vector endpoints projected onto the vertical and horizontal planes, respectively. T and M indicate thermal (°C) and AF (mT) demagnetization, respectively.

(AF) demagnetization. The hybrid thermal–AF technique (i.e. AF after thermal demagnetization) was used for specimens with a hard viscous/secondary remanent magnetization that was not appreciably removed by AF treatment alone and showing unstable behavior during thermal demagnetization associated with peak temperatures above 300°C. Thermal demagnetization was carried out with a Natsuhara TSD-1 demagnetizer unit, with steps as large as 100°C (lower temperatures) to 10°C (higher temperatures). The hybrid thermal–AF demagnetization was carried out as a progressive thermal demagnetization up to 250 or 300°C followed by AF demagnetization at 5 and 20 mT steps at lower and higher peak fields, respectively, using a Natsuhara DEM-8601C demagnetizer. Magnetic remanence measurements were performed on a 2G cryogenic magnetometer in the paleomagnetic laboratory at Kobe University in Japan. Demagnetization results were plotted on orthogonal diagrams [18] and stereographic projections. Directions of magnetization were determined by principal component analysis [19] and calculated at each site using standard Fisher [20] statistics. When appropriate the method of combined remagnetization circles and direction observations of McFadden and McElhinny [21] was employed.

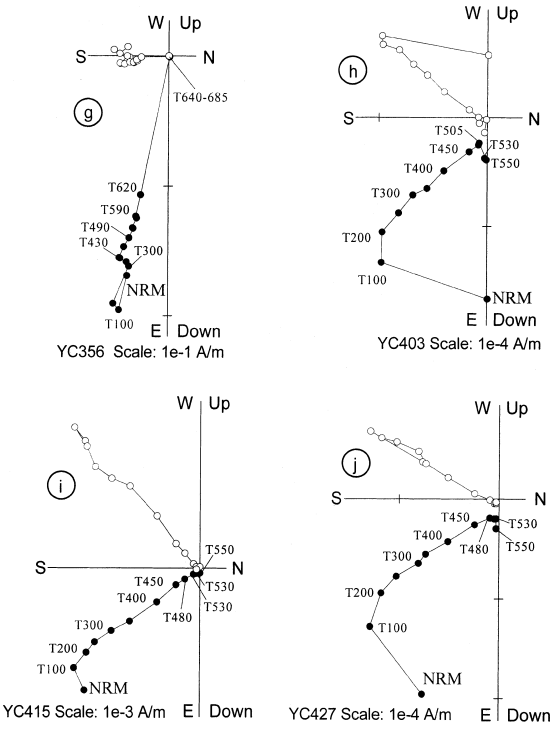


Fig. 3 (*continued*).

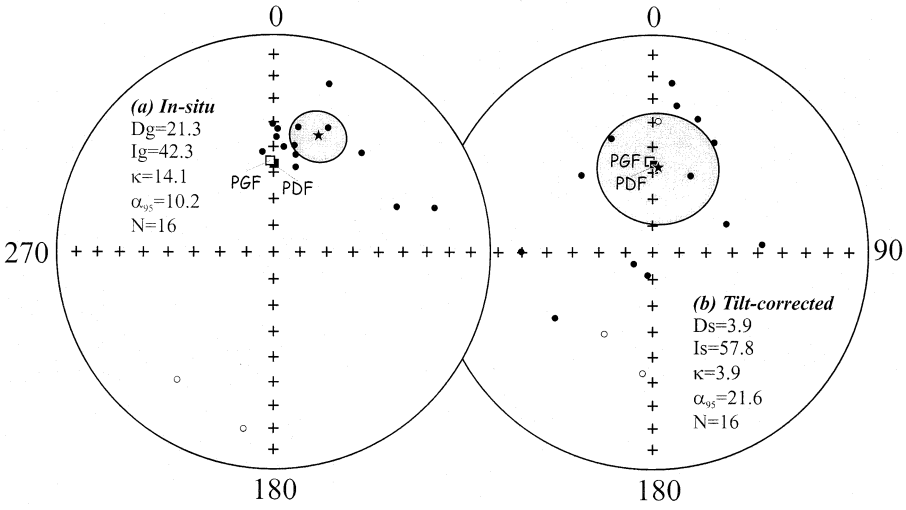


Fig. 4. Equal-area projection of site-mean directions for the A-component from the Carboniferous before (a) and after (b) tilt correction. Solid/open circles represent directions plotted onto the lower/upper hemisphere. Stars represent mean directions. PGF and PDF indicate present geomagnetic field and present dipole field, respectively.

3.2. Low-temperature component

After removal of recent viscous overprint components at 100 or 200°C, the magnetic remanence of most specimens was found to be comprised of two magnetic components (Fig. 3), a low-temperature and a high-temperature component. The contribution of the viscous overprint components to the NRM was variable but for the Early Carboniferous specimens can represent more than 50% of the total intensity. The low-temperature A-component with mostly north-by-northeast and moderate to steeply downward dipping in-situ directions was removed at peak demagnetization temperatures below 300 or 350°C. This component has an in-situ mean direction of $D = 21.3°$, $I = 42.3°$, $\alpha_{95} = 10.2°$, $N = 76$ specimens, and a tilt-corrected direction of $D = 3.9°$, $I = 57.8°$, $\alpha_{95} = 21.6°$ (Table 1, Fig. 4). Directions for this component are mostly normal, with a few specimens (eight) showing reverse polarity (Table 1). The fold test [22] is distinctly negative at the 95% confidence limit (in-situ $\xi_1 = 3.317$, tilt-corrected $\xi_1 = 4.76$, $\xi = 4.650$, critical value at 95% confidence), and the significant decrease in the precision parameter after tilt correction (from 14.1 to 3.9; $\kappa_g/\kappa_s = 3.62$) suggests the A-component to be of post-folding origin.

3.3. High-temperature component

Following the removal of the viscous and low-temperature A-component, a high-temperature B-component was generally isolated from the limestone specimens between 350 and 530–550°C (Fig. 3b,h–j), and was usually isolated after demagneti-

Table 2
Summary of the high-temperature B-component results from the Hexi Corridor

Site ID	Strike	Dip	n/n_0	R/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95}
Early–Middle Carboniferous, limestone and one site of sandstone (YC35):										
YC28	128	74	5/5	5/0	135.2	−28.4	102.0	−13.7	22.1	16.6
YC29	282	83	0/3	No high-temperature component isolated						
YC30	139	66	2/5	2/0	123.5	−42.8	94.3	−5.6	–	–
YC31	21	20	2/2	2/0	97.6	11.6	97.8	−8.0	–	–
YC32	326	20	4/8	4/0	98.1	−7.1	101.7	−21.6	84.9	10.1
YC33	154	37	6/9	6/0	108.3	−38.6	97.6	−9.3	29.6	12.5
YC34	103.4[a]	45.9[a]	6/7	5/1	104.8	−18.9			17.2	16.6
							90.5	−14.0	48.9	9.7
YC35	115	26	2/8	3/0	98.5	0.8	100.4	7.8	132.5	10.8
YC36	139	28	0/8	No high-temperature component isolated						
YC37[b]	139	28	4/6	4/0	126.8	−10.4	123.4	−3.5	153.7	10.1
YC45	119	59	0/8	No high-temperature component isolated						
YC46[b]	131	31	2/5	2/0	103.6	−28.6	91.6	−16.0	–	–
YC47	265	27	6/7	6/0	84.8	−19.4	93.9	−17.3	25.1	13.6
Averages			10/13	39/1	107.4	−18.8			13.1	13.9
							99.3	−10.3	42.3	7.5
Late Carboniferous, limestone:										
YC40	161	22	5/5	5/0	129.2	−52.6	112.0	−38.1	18.4	18.3
YC41[b]	161	22	3/7	3/0	149.6	−37.3	135.5	−30.2	262.2	12.6
YC42	161	23	6/6	6/0	140.1	−34.8	128.3	−24.2	44.8	10.1
YC43	163	21	2/2	2/0	111.6	−39.1	104.5	−21.8	–	–
YC44	163	21	0/7	No high-temperature component isolated						
Averages			4/5	16/0	133.1	−41.8			30.1	17.0
							120.1	−29.1	31.2	16.7

[a] Mean value (YC34a (340–341), strike/dip = 120/48; YC34b (342–347), strike/dip = 90/47).
[b] Site-mean observation was calculated by combining remagnetization great circles with direction observations using McFadden and McElhinny's method [21]. See Table 1.

zation to peak AFs of 20–30 mT (Fig. 3a,c–f). In some sites, however, the low-temperature A-component almost obliterated the high-temperature B-component, preventing its separation (Table 2). Demagnetization of some specimens from three sites, YC37, YC41 and YC46, yielded a distinct great-circle trend toward the characteristic direction; in these cases the site-mean direction for the B-component was calculated by combining remagnetization great circles with direction observations [21]. A few sandstone samples were collected for comparison of magnetization components between limestone and sandstone rocks. Unfortunately, an erratic demagnetization behavior was revealed for most sandstone specimens; only three of eight specimens yielded a high-temperature B-component between peak temperatures of 450 to 680°C (Fig. 3g).

In general, the direction of the B-component, isolated from most Carboniferous specimens, clusters after tilt correction with a southeast-by-east shallow upward inclination. All sites but four provided consistent high-temperature B-component directions with exclusively reverse polarity for either the Early-Middle Carboniferous or Late Carboniferous (normal polarity was isolated from only one specimen from site YC34 (Fig. 3f)). Site-mean directions for this component are listed in Table 2 and shown in Fig. 5. The B-component has a tilt-corrected mean direction of $D = 99.3°$, $I = -10.3°$, $\kappa = 42.3$, $\alpha_{95} = 7.5°$, $N = 10$ sites, for the Early-Middle Carboniferous, and of $D = 120.1°$, $I = -29.1°$, $\kappa = 31.2$, $\alpha_{95} = 16.7°$, $N = 4$ sites, for the Late Carboniferous (Table 2 and Fig. 5b,d). The mean direction for the Late Carboniferous B-component has a large uncertainty ($\alpha_{95} > 15°$). We have no definite explanation for this except, perhaps, that the number of specimens/sites involved in the calculation is small. As shown in Fig. 5a,b, clustering of the B-component directions for the Early-Middle Carboniferous specimens is strikingly improved after com-

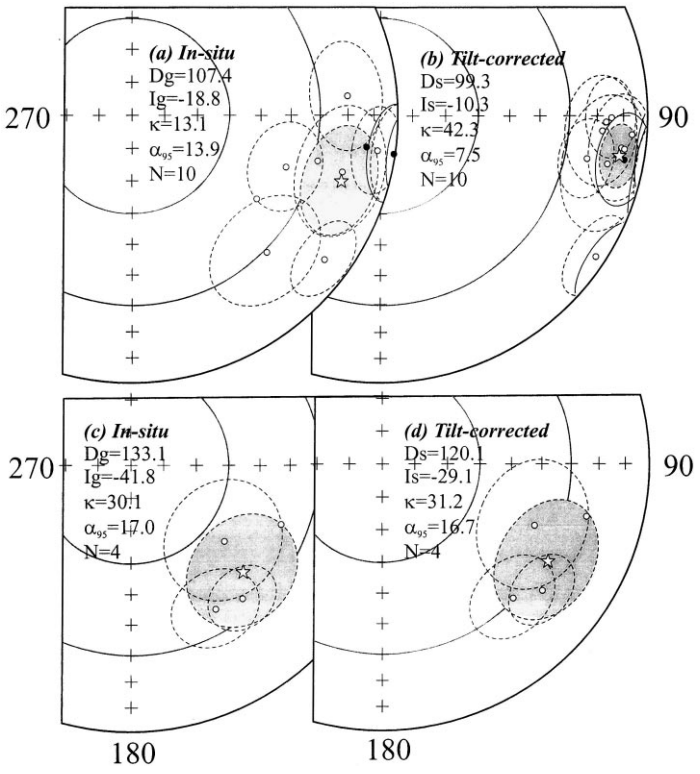


Fig. 5. Equal-area projection of site-mean directions of the B-component from the Early-Middle (a, b) and Late (c, d) Carboniferous before (a, c) and after (b, d) tilt correction. See Fig. 4.

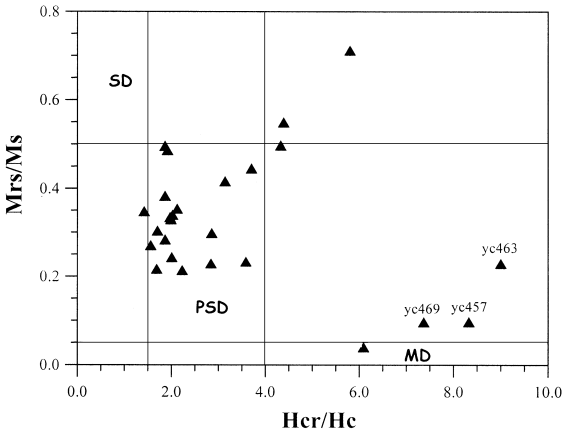


Fig. 6. Day diagram [23] of some limestone specimens indicating the domain remanence carriers to be within PSD range. The hysteresis loop parameters were observed using a Princeton/MicroMag model 2900 alternating gradient magnetometer with a maximum applied field of 1.5 T. Three specimens showing a high ratio of H_{cr}/H_c were labelled by specimen number.

plete unfolding ($\kappa_s/\kappa_g = 3.23$). The fold test [22] is positive at the 95% confidence level for these Early-Middle Carboniferous specimens (in-situ $\xi_1 = 3.698$, tilt-corrected $\xi_1 = 1.207$, $\xi = 3.685$, critical value at 95% confidence for $N = 10$). No fold test is possible for Late Carboniferous specimens, since they were collected from a monocline. However, the fold test [22] is positive at the 95% confidence level for all Early-Middle and Late Carboniferous specimens (in-situ $\xi_1 = 5.075$, tilt-corrected $\xi_1 = 1.692$, $\xi = 4.20$, critical value at 95% confidence for $N = 14$). Consequently, the B-component of all Carboniferous specimens is of pre-folding i.e. pre-Early-Middle Jurassic origin.

3.4. Rock magnetism

The demagnetization behavior and rock magnetic results (performed on a Princeton/MicroMag model 2900 alternating gradient magnetometer at the paleomagnetic laboratory of the Institute of Geology and Geophysics of Chinese

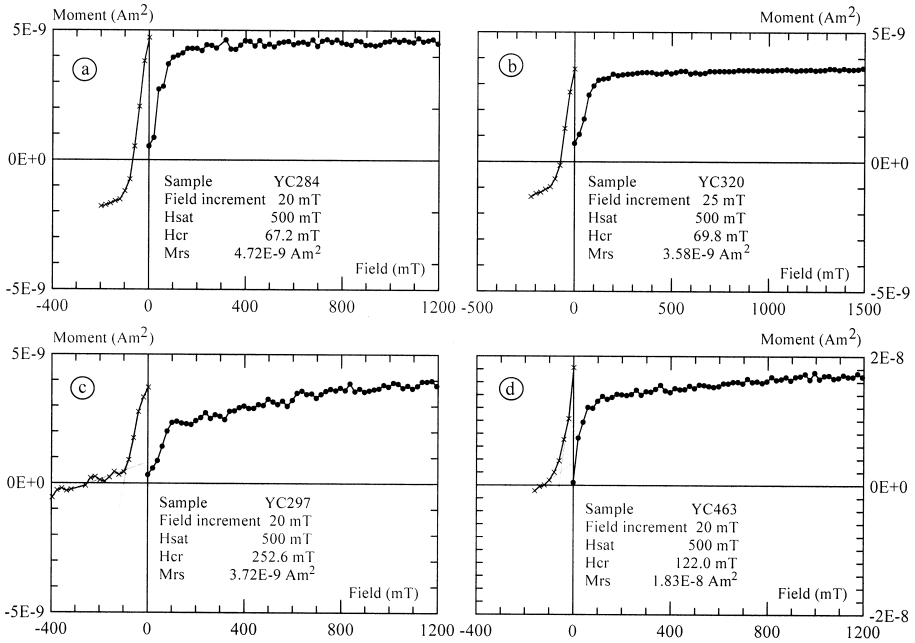


Fig. 7. Behavior of acquisition and back-demagnetization of IRM for representative Carboniferous limestone specimens. All measured on a Princeton/MicroMag model 2900 alternating gradient magnetometer. Solid circles and crosses indicate IRM acquisition and back-demagnetization of IRM acquired in a maximum applied field (H_{sat}) of 500 mT, respectively.

Academy of Sciences) for the most of the limestone rocks, suggest that the B-component may reside in pseudo-single domain (PSD) magnetite. The Day plot [23] of hysteresis loop parameters (Fig. 6) for some limestone specimens shows most carriers of the remanence to be within the PSD range. Acquisition of isothermal remanent magnetization (IRM) and back-demagnetization experiments (Fig. 7a,b) shows saturation typically to occur by 0.3 or 0.5 T with remanent coercivities (H_{cr}) generally less than 0.1 T. These results indicate magnetite as the predominant carrier of remanence in these limestone specimens. Some fraction of high-coercivity magnetic minerals such as hematite, however, is also found to be present. As was the case in our previously published IRM results [12], IRM acquisition is also characterized by a quasi-saturation behavior at about 0.2 T and a further gradual increase in intensity up to 1.2 T (Fig. 7c,d). At the same time, back-demagnetization experiments find these specimens with high coercivity, however, a significant trend to relative lower remanent coercivity can also be found in the initial stage of back-demagnetization (Fig. 7c,d). Rocks having an appearance of a high-coercivity magnetic fraction usually show a high ratio of H_{cr}/H_c (Fig. 6, labelled three specimens), and generally no high-temperature components were isolated from these specimens.

4. Discussion

4.1. Origin of the components of magnetization

The fold test indicates that the B-component was acquired before folding whereas the A-component was recorded after folding with the relevant folding to be of Early to Middle Jurassic age. The corresponding in-situ paleomagnetic pole of the A-component, at 224.3°E, 67.8°N, with semi-axis along and perpendicular to the great circle of ellipse $dp/dm = 7.7/12.6$, is not statistically distinguishable from those of the Middle-Late Jurassic or Early Cretaceous for the NCB [24,25] at the 95% confidence limit (Fig. 8), indicating that the A-component is probably a Late Mesozoic over-

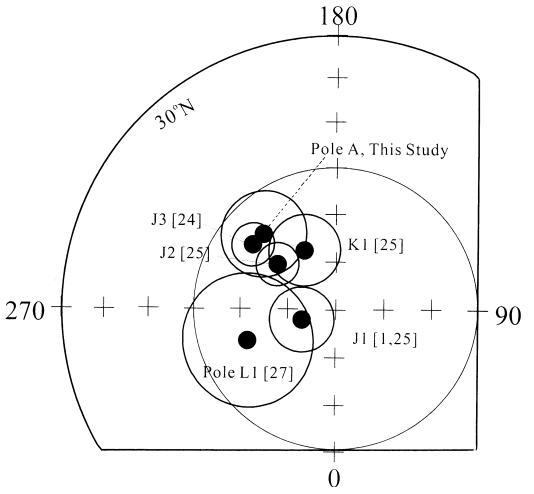


Fig. 8. Equal-area projection showing comparison between remagnetized Pole A (corresponding to the A-component) and Pole L1 (calculated from sites T5–T10 of [27]) and some available Jurassic (J) and Cretaceous (K) poles for the NCB [1,24,25]. All the poles are plotted onto the Northern Hemisphere with 95% confidence limit.

print. We note that this overprint is pervasive and clearly defined in the Paleozoic rocks from the eastern part of the Hexi Corridor, regardless of rock age or lithology.

Together with the pre-Jurassic age of the B-component, the exclusively reversed polarity indicates that the characteristic high-temperature B-component was acquired earlier than the mixed polarity Triassic and probably during the reverse polarity Permo-Carboniferous Superchron, and this would place the age of acquisition of the magnetization before about 263 Ma [26]. Two corresponding paleomagnetic poles, which are at 14.0°E, 10.5°N, $\kappa = 62.4$, $\alpha_{95} = 6.2°$ and 10.2°E, 33.3°N, $\kappa = 31.3$, $\alpha_{95} = 16.7°$ respectively, are significantly different from the post-Permian paleopoles for the NCB (Fig. 9), and could be representative for the Carboniferous.

On the other hand, the significant difference between our late Visean/Namurian pole and that for the Stephanian may indicate that our late Visean/Namurian results were acquired before the Stephanian. These observations and their presence in different lithologies imply that the characteristic high-temperature B-component was acquired at, or close to, the time of formation of the rocks.

4.2. Tectonic and paleogeographic implications

There are only a few Carboniferous results reported for the NCB, namely from the Middle-Late Carboniferous sandstone and limestone rocks of Shanxi Province [27], and from the Late Carboniferous Taiyuan Formation of Shaanxi Province [28]. The results of the former study appear to show a secondary magnetization at some sites. Indeed, six out of eight sites (T5–T10, [27]) yield a mean paleomagnetic pole at 289.2°E, 71.0°N, $\alpha_{95} = 13.9°$, which is in agreement with an Early Jurassic pole of the NCB [1] at the 95% confidence limit (Fig. 8), suggesting that observations from these sites probably represent an Early Jurassic magnetization. The remaining two sites (T4 and T11) have reverse polarity and yield a paleomagnetic pole at 28.0°E, 25.0°N. This pole is significantly different from the post-Permian poles of the NCB (Fig. 9), suggesting that it may be representative of the Middle-Late Carboniferous. The pole of the latter study (12.0°E, 30.0°N, $\alpha_{95} = 6.1°$) shows a positive fold test, but is based on a limited number of specimens (15) and was analyzed using a more classical technique without principal component analysis. It was, therefore, not considered during construction of the Phanerozoic APW path for the NCB [1]. Our Late Carboniferous pole, however, is in agreement with these earlier determined poles at the 95% confidence (Fig. 9) and, hence, is better to support the contention that the Hexi Corridor terrane had been the western extension of the NCB since Ordovician time [15].

Our two newly determined Carboniferous paleomagnetic poles are distinct from coeval ones for Inner Mongolia [25], Outer Mongolia [29], South China [5], Tarim [30–33] and stable Europe

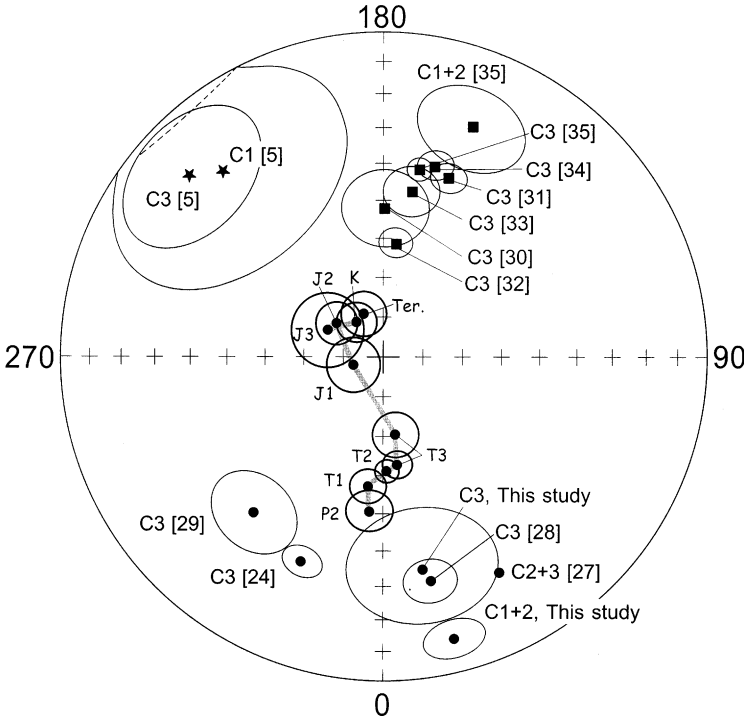


Fig. 9. Equal-area projection of a post-Late Permian APW path for the NCB (circles) and Carboniferous poles for Tarim and Siberia (squares), South China (stars), and Inner Mongolia, Outer Mongolia, and the NCB (circles). The APW path for the NCB is drawn according to Yang et al. [1]. Paleomagnetic poles of Irving and Irving [34] for 'stable' Europe, and Lin et al. [27] for the NCB are drawn after re-analysis of Zhao et al. [10] and this study, respectively. C, P, T, and Ter. indicate Carboniferous, Permian, Triassic and Tertiary, respectively. See Fig. 8.

[34,35] at greater than 95% confidence (Fig. 9). These results suggest that these continental blocks and the NCB could not have been connected in their present configuration during Carboniferous time.

4.2.1. Early-Middle Carboniferous

Comparison of our Early-Middle Carboniferous pole with Silurian and Devonian data from the eastern part of the Hexi Corridor terrane [13], suggests that a significant angular distance of $53.8° \pm 8.9°$ is expected for the NCB from the Middle-Late Devonian through late Early Carboniferous. This motion reflects the tectonic motion of the NCB during that period: clockwise rotation of $53.6° \pm 9.0°$ accompanied by slight southward latitudinal translation of $12.0° \pm 8.9°$. A similar trend of clockwise rotation and southerly drift is also observed in the South China block (SCB) and Australia of eastern Gondwanaland (Table 3), indicating that both the NCB and SCB were significantly affected by the rapid southward drift of Gondwanaland during that period. However, the different amount of movement and the distinct difference in paleolatitude between the two blocks of China and Australia of eastern Gondwanaland (Table 3 and Fig. 10) suggest that the NCB and SCB had separated together from Gondwanaland by the late Visean. Given the premise that the NCB and SCB had been located close to the Australian part of Gondwanaland during the Late Devonian [10,13,16], we now present an Early Carboniferous paleogeographic reconstruction of the NCB and those regions of the world adjacent to it (Fig. 10a). The NCB and SCB may have had similar latitudinal positions, and may not have been separated by a great distance. Nevertheless, the observation of distinct Carboniferous paleomagnetic poles (Fig. 9) indicates that these two blocks could not have been directly connected during the Early-Middle Carboniferous.

4.2.2. Late Carboniferous

Our Late Carboniferous pole makes possible a preliminary paleogeographic reconstruction of the NCB in a global setting during that period. The NCB and SCB seem to have remained in proximity of the paleo-equator in an eastward opening ocean (Fig. 10b). In the period between the late Visean–Namurian and Stephanian, angular distance of $23.1° \pm 14.3°$ indicates that the NCB was mainly subjected to a counterclockwise rotation of $20.2° \pm 14.5°$, accompanied by no/slight northerly latitudinal drifting ($11.5° \pm 14.3°$). On the contrary, the SCB, seemingly influenced by

Table 3
Comparison of Late Paleozoic poles for Australia, the North and South China blocks

Age	Pole position, A_{95}	Paleolatitude	$R \pm \Delta R$	$F \pm \Delta F$	Ref.
Australia (reference site: 15°S, 135°E)[a]*:*					
D3	217.5E, 56.5N	8.4°S			[35]
C1	272.9E, 73.4N	26.9°S	$-21.2° \pm ?$	$-18.5° \pm ?$	[35]
C3	301.5E, 51.5N, 9.0	51.9°S	$1.2° \pm ?$	$-25.0° \pm ?$	[36]
P2	338.0E, 47.4N, 7.0	52.4°S	$-39.3° \pm 14.9°$	$-0.5° \pm 9.1°$	[36]
North China block (reference site: 35°N, 119°E):					
D2+3	337.0E, 57.0N, 9.2	7.4°N			[13]
C1+2	14.0E, 10.5N, 6.2	6.0°S	$-53.6° \pm 9.0°$	$-12.0° \pm 8.9°$	this study
C3	10.2E, 33.3N, 16.7	5.4°N	$20.2° \pm 14.5°$	$11.5° \pm 14.3°$	this study
P2	355.2E, 50.3N, 5.7	8.6°N	$20.6° \pm 14.1°$	$2.6° \pm 14.1°$	[1]
South China block (reference site: 35°N, 119°E)					
S2+3	195.7E, 6.8N, 5.3	14.8°N			[37]
C1	220.7E, 25.2N, 28.2	5.4°N	$-29.1° \pm 23.0°$	$-9.4° \pm 23.0°$	[5]
C3	227.1E, 19.4N, 16.9	2.8°S	$1.0° \pm 26.4°$	$-8.2° \pm 26.3°$	[5]
P2	243.6E, 55.3N, 5.2	10.5°N	$-35.2° \pm 14.2°$	$14.8° \pm 14.1°$	[2]

Mean paleopoles are given in their own coordinate system. R and F indicate rotation (+/−, counterclockwise/clockwise) and paleo-latitudinal difference (+/−, northward/southward), respectively.

[a] Mean Gondwana paleopoles [35,36] are used. The rotation parameters are from Lottes and Rowley [38].

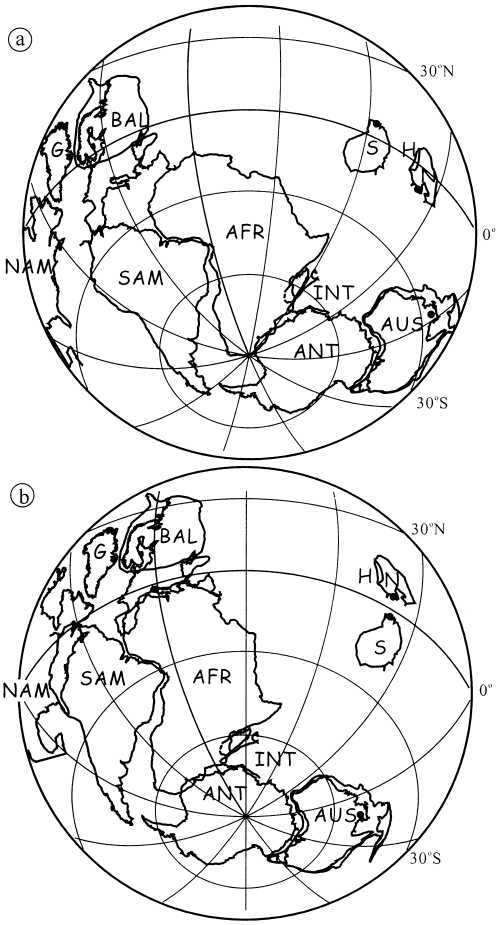


Fig. 10. Schematic maps showing paleogeographic reconstruction of the NCB in a global setting during (a) the Early and (b) Late Carboniferous. Early Carboniferous Gondwanaland and Laurussia are shown according to Van der Voo [35], whereas Late Carboniferous Gondwanaland and Laurussia are shown according to Late Carboniferous mean poles of [36] with G_3 = the octupole/dipole field ratio = 0 (the without Australia pole is used). The rotation parameters for Gondwanaland and Laurussia are from Lottes and Rowley [38] and Torsvik et al. [39], respectively. The SCB is shown according to its Carboniferous poles of Lin et al. [5]. Relative longitudes of the blocks are unconstrained. Abbreviations are: AFR, Africa; ANT, Antarctica; AUS, Australia; BAL, Baltica; G, Greenland; H; Hexi Corridor; IND, India; NAM, North America; NCB, North China block; SAM, South America; SCB, South China block. Closed circles indicate reference sites used in Table 3.

the rapid southerly drift of Gondwanaland, shows a continuous southward drift during that period (Table 3). The different tectonic motions of the NCB and SCB during the period between the Early-Middle and Late Carboniferous apparently lead to a longitudinal juxtaposition of the two blocks, separated somewhat in latitude during the Late Carboniferous (Fig. 10b). Indeed, the geochronological coherence between the paleomagnetism and the metamorphism favors strongly the suggestion that the NCB and SCB had reached similar longitudinal positions by the Late Permian. The comparison between the NCB and SCB APW paths from the Late Permian to Late Triassic clearly shows that the two tracks follow similar trends, indicating minor relative motion during this period [40]. This period is then followed by a scissor-like collision process [6] of the two blocks with a fast counterclockwise rotation of the NCB with respect to the SCB. The two blocks may have already sutured during the Middle Jurassic and may be even as early as Early Jurassic, in depending on statistic similarity of Middle Jurassic pole of the NCB to those of the Early-Middle Jurassic of the SCB [40]. On the other hand, the strong magmatic activity around 220–210 Ma in the western Qinling, and the occurrence of ultra high-pressure coesite eclogites dated at 235–205 Ma by Sm/Nd and U/Pb methods and at 228–205 Ma by Ar/Ar dating in the eastern Qinling (Dabie-Shan area) also suggest important geodynamic events related to the NCB-SCB relative motions and subsequent collision during the Late Triassic or earliest Jurassic ([40] and reference therein).

Of course, Carboniferous global reconstructions and tectonic implications of our new pole positions presented in this paper heavily depend on the Hexi Corridor-NCB relationship and on the veracity of the late Paleozoic APW path for Gondwanaland.

5. Conclusions

1. Late Carboniferous paleomagnetic pole from the eastern part of the Hexi Corridor, being not significantly different from earlier, provisional findings for the stable NCB, reinforce

the claim that the Hexi Corridor terrane had been the western extension of the NCB since Ordovician time.

2. Two Carboniferous paleomagnetic poles from the eastern part of the Hexi Corridor are distinct with the coeval ones for Inner Mongolia, Outer Mongolia, South China, Tarim and stable Europe, at greater than 95% confidence, suggesting that these continental blocks could not have been connected in their present configuration during the Carboniferous.

3. Significantly different paleolatitudes and tectonic motions between the NCB and eastern Gondwanaland during the Carboniferous indicate that the two had rifted apart by the late Visean.

Acknowledgements

We are very grateful to the Geological Bureau of Ningxia Hui Autonomous Region and Lixin Bai for organizing the field trip and to Jianjun Lu for his participation and help in the field. We wish to thank C. Klootwijk and P. Schmidt for their careful reviews and comments. We also acknowledge K. Hoffman for providing constructive review of the manuscript. This study was supported by the NSF of China (No. 49874014) and the CAS (KZCX1-07), and partially supported by the Ministry of Personnel of China. Paleomagnetic data were analyzed using R. Enkin's, T. Torsvik's and J.P. Cogné's computer program packets.*[RV]*

References

[1] Z.Y. Yang, X.H. Ma, B.C. Huang, Z.M. Sun, Y.X. Zhou, Apparent polar wander path and tectonic movement of the North China Block in Phanerozoic, Sci. China Ser. D 41 (Suppl.) (1998) 51–65.
[2] H.N. Wu, R.X. Zhu, L.X. Bai, B. Guo, J.J. Lu, Revised apparent polar wander path of the Yangtze block and its tectonic implications, Sci. China Ser. D 41 (Suppl.) (1998) 78–90.
[3] D.J. Fang, P.Y. Wang, Z.Y. Shen, X.D. Tan, Cenozoic paleomagnetic results and Phanerozoic apparent polar wandering path of Tarim block., Sci. China Ser. D 41 (Suppl.) (1998) 105–112.
[4] K.N. Huang, N.D. Opdyke, R.X. Zhu, Further paleomagnetic results from the Silurian of the Yangtze block and their implications, Earth Planet Sci. Lett. 175 (2000) 191–202.
[5] J.L. Lin, M. Fuller, W.Y. Zhang, Preliminary Phanerozoic polar wander paths for the North and South China blocks, Nature 313 (1985) 444–449.
[6] X.X. Zhao, R.S. Coe, Paleomagnetic constraints on the collision and rotation of North and South China, Nature 327 (1987) 141–144.
[7] Y.J. Zhai, Y.X. Zhou, Paleomagnetism and tectonic evolutions of North and South China blocks since the Phanerozoic, Acta Geophys. Sin. 32 (1989) 292–307. (in Chinese)
[8] Y.P. Li, An apparent polar wander path from the Tarim block, China, Tectonophysics 181 (1990) 31–41.
[9] R.J. Enkin, Z.Y. Yang, Y. Chen, V. Courtillot, Paleomagnetic constrains on the geodynamic history of the major blocks of China from the Permian to the Present, J. Geophys. Res. 97 (1992) 13953–13989.
[10] X.X. Zhao, R.S. Coe, S.A. Gilder, G.M. Frost, Paleomagnetic constrains on the paleogeography of China: implications for Gondwanaland, Aust. J. Earth Sci. 43 (1996) 643–672.
[11] L.X. Bai, R.X. Zhu, H.N. Wu, B. Guo, J.J. Lu, New Cambrian paleomagnetic pole for Yangtze block, Sci. China Ser. D 41 (Suppl.) (1998) 66–71.
[12] B.C. Huang, Z.Y. Yang, Y. Otofuji, R.X. Zhu, Early Paleozoic paleomagnetic poles from the western part of North China block and their implications, Tectonophysics 308 (1999) 377–402.
[13] B.C. Huang, Y. Otofuji, Z.Y. Yang, R.X. Zhu, New Silurian and Devonian palaeomagnetic results from the Hexi Corridor terrane, Northwest China and their tectonic implications, Geophys. J. Int. 140 (2000) 132–146.
[14] H.N. Wu, L.F. Zhou, Z.Y. Zhao, Z.Y. Yang, Y. Chen, Tectonic implications of the paleomagnetic results of the later Paleozoic and Mesozoic rocks from the Alashan area of the western North China block, Sci. Geol. Sin. 2 (1993) 19–46.
[15] B.C. Huang, Y. Otofuji, Z.Y. Yang, Paleomagnetic constraints on the tectonic relationship between the Alashan/Hexi Corridor terrane and the North China Block, Geophys. Res. Lett. 26 (1999) 787–790.
[16] X.X. Zhao, R.S. Coe, H.N. Wu, Z.Y. Zhao, Silurian and Devonian paleomagnetic poles from North China and implications for Gondwana, Earth Planet Sci. Lett. 117 (1993) 497–506.
[17] Ningxia Bureau of Geology and Mineral Resources, Regional Geology of Ningxia Hui Autonomous Region, Geol. Publ. House, Beijing, 1990, p. 552.
[18] J.D.A. Zijderveld, AC demagnetization of rocks: Analysis of results, in: D.W. Collinson, K.M. Creer, S.K. Runcorn

(Eds.), Methods on Paleomagnetism, Elsevier, Amsterdam, 1967, pp. 254–286.
[19] J.L. Kirschvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[20] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. Lond. A217 (1953) 295–305.
[21] P.L. McFadden, M.W. McElhinny, The combined analysis of remagnetization circles and direct observations in palaeomagnetism, Earth Planet Sci. Lett. 87 (1988) 161–172.
[22] P.L. McFadden, A new fold test for palaeomagnetic studies, Geophys. J. Int. 103 (1990) 163–169.
[23] R. Day, M. Fuller, V.A. Schmidt, Hysteresis properties of titanomagnetite: grain-size and compositional dependence, Phys. Earth Planet. Inter. 13 (1977) 260–266.
[24] X.X. Zhao, R. Coe, Y.X. Zhou, H.R. Wu, J. Wang, New paleomagnetic results from northern China, collision and suturing with Siberia and Kazakhstan, Tectonophysics 181 (1990) 43–81.
[25] Z.Y. Yang, V. Courtillot, J. Besse, X. Ma, L. Xing, S. Xu, J. Zhang, Jurassic paleomagnetic constraint on the collision of the North and South China Blocks, Geophys. Res. Lett. 19 (1992) 577–580.
[26] N.D. Opdyke, J.E.T. Channell, Magnetic Stratigraphy, Academic Press, San Diego, CA, 1996, p. 346.
[27] W.Z. Lin, J. Shao, Z.Y. Zhao, Paleomagnetic features of Sino-Korean plate in Late Paleozoic era, Geophys. Geochem. Explor. 8 (1984) 297–304. (in Chinese with English abstract)
[28] H. Wu, Apparent polar wander paths and paleolatitude distributions for the North and South China blocks, the geotectonic evolution of Qinling belt, Ph.D thesis, Institute of Geology, Academia Sinica, Beijing, 1988, p. 175 (in Chinese).
[29] P. Pruner, Paleomagnetism and paleogeography of Mongolia from the Carboniferous to the Cretaceous: final report, Phys. Earth Planet. Inter. 317 (1992) 500–505.
[30] Y. Bai, G. Chen, Q. Sun, Y.A. Li, Y. Dong, D. Sun, Late Paleozoic polar wander path for the Tarim platform and its tectonic significance, Tectonophysics 139 (1987) 145–153.
[31] Y. Zhai, Y. Li, M. McWilliams, Late Carboniferous paleomagnetism of the Tarim craton China (abstract), EOS Am. Geol. Union Trans. 68 (1987) 1259.
[32] Z.F. Meng, J. Zheng, Paleomagnetic investigations of the apparent polar wander path of Tarim and relative motions between sub-tectonic units, in: Q. Zhou, J. Zheng (Eds.), Tectonic Synthesis of Tarim, Science Press, Beijing, 1990, pp. 50–87.
[33] D. Fang, G. Jin, L. Jiang, P. Wang, Z. Wang, Paleomagnetic results and the tectonic significance of Tarim plate, Acta Geophys. Sin. 39 (1996) 522–532. (in Chinese)
[34] E. Irving, G.A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwanaland, Geophys. Surv. 5 (1982) 141–188.
[35] R. Van der Voo, Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans, Cambridge University Press, New York, 1993, p. 441.
[36] R. Van der Voo, T.H. Torsvik, Evidence for late Paleozoic and Mesozoic non-dipole fields provides an explanation for the Pangea reconstruction problems, Earth Planet. Sci. Lett. 187 (2001) 71–81.
[37] N.D. Opdyke, K. Huang, G. Xu, W.Y. Zhang, D.V. Kent, Paleomagnetic results from the Silurian of the Yangtze paraplatform, Tectonophysics 139 (1987) 123–132.
[38] A.L. Lottes, D.B. Rowley, Reconstruction of the Laurensian and Gondwana segments of Permian Pangaea, in: W.S. McKerrow, C.R. Scotese (Eds.), Palaeozoic Palaeogeographic and Biogeography, vol. 12, Geol. Soc. London Memoir, 1990, pp. 383–395.
[39] T.H. Torsvik, R. Van der Voo, J.G. Meert, J. Mosar, H.J. Walderhaug, Reconstruction of the contients around the North Atlantic at about the 60th parallel, Earth Planet. Sci. Lett. 187 (2001) 55–69.
[40] Z.Y. Yang, J. Besse, New Mesozoic apparent polar wander path for south China: tectonic consequences, J. Geophys. Res. 106 (2001) 8493–8520.